

React JS Hands-on Lab

Function Components & Styling – Student Score Calculator App

Objective

Create a React app for a Student Management Portal named **scorecalculatorapp** and create a function component named **CalculateScore** which accepts Name, School, Total and goal in order to calculate the average score of a student and display the same.

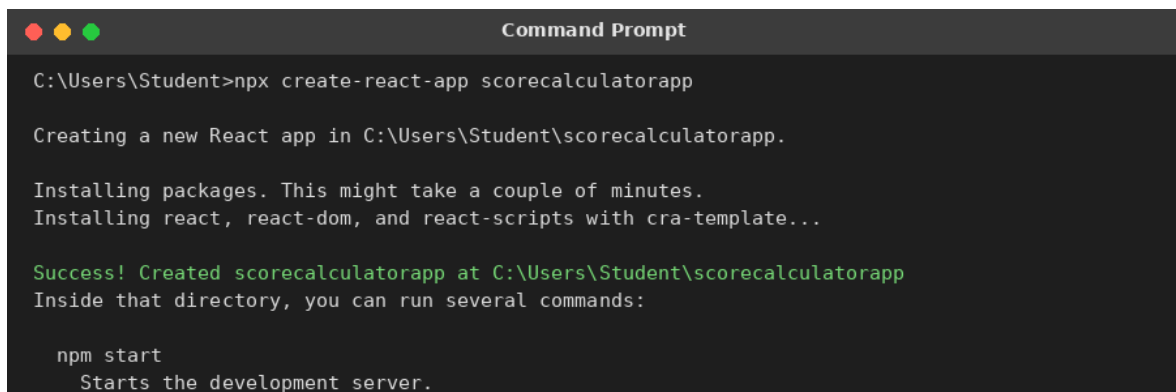
In this lab I have learnt how to:

- Create a function component
- Apply style to components
- Render a component

Step 1: Create the React Project

Opened the terminal in Visual Studio Code and created a new React project named **scorecalculatorapp** using the following command:

```
npx create-react-app scorecalculatorapp
```



```
Command Prompt

C:\Users\Student>npx create-react-app scorecalculatorapp

Creating a new React app in C:\Users\Student\scorecalculatorapp.

Installing packages. This might take a couple of minutes.
Installing react, react-dom, and react-scripts with cra-template...

Success! Created scorecalculatorapp at C:\Users\Student\scorecalculatorapp
Inside that directory, you can run several commands:

  npm start
    Starts the development server.
```

Screenshot: Output of create-react-app command in terminal

Step 2: Create the CalculateScore Component

Created a new folder named **Components** under the **src** folder and added a new file named **CalculateScore.js** with the following code:

```
import '../Stylesheets/mystyle.css';

const percentToDecimal = (decimal) => {
  return (decimal.toFixed(2) + '%');
}

const calcScore = (total, goal) => {
  return percentToDecimal(total / goal);
}

export const CalculateScore = ({Name, School, total, goal}) => (
  <div className="formatstyle">
```

```

    <h1><font color="Brown">Student Details:</font></h1>
    <div className="Name">
      <b><span>Name: </span></b>
      <span>{Name}</span>
    </div>
    <div className="School">
      <b><span>School: </span></b>
      <span>{School}</span>
    </div>
    <div className="Total">
      <b><span>Total: </span></b>
      <span>{total}</span>
      <span> Marks</span>
    </div>
    <div className="Score">
      <b>Score: </b>
      <span>
        {calcScore(total, goal)}
      </span>
    </div>
  </div>
)

```

Code: src/Components/CalculateScore.js

Step 3: Add Styling to the Component

Created a folder named **Stylesheets** and added a file named **mystyle.css** to style the component:

```

.Name
{
  font-weight: 300;
  color: blue;
}

.School
{
  color: crimson;
}

.Total
{
  color: darkmagenta;
}

.formatstyle
{
  text-align: center;
  font-size: large;
}

.Score
{
  color: forestgreen;
}

```

Code: src/Stylesheets/mystyle.css

Step 4: Render the Component in App.js

Edited **App.js** to import and invoke the CalculateScore functional component:

```
import './App.css';
import { CalculateScore } from './Components/CalculateScore';

function App() {
  return (
    <div>
      <CalculateScore
        Name={"Steeve"}
        School={"DAV Public School"}
        total={284}
        goal={3}
      />
    </div>
  );
}

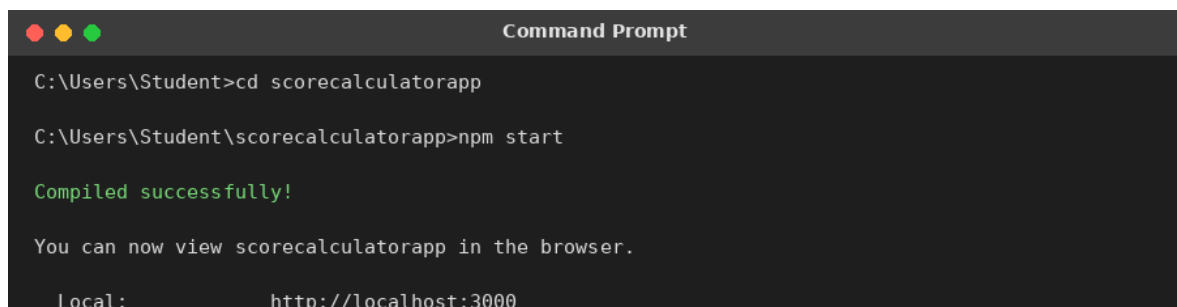
export default App;
```

Code: src/App.js

Step 5: Run the Application

In the command prompt, navigated into the scorecalculatorapp folder and started the development server:

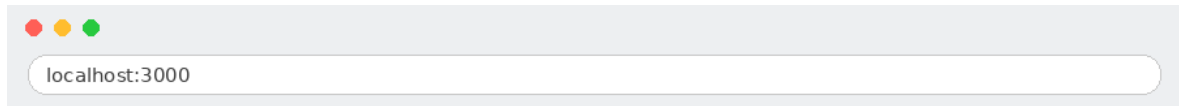
```
cd scorecalculatorapp
npm start
```



Screenshot: Output of npm start command in terminal

Step 6: View the Output in Browser

Opened the browser and navigated to **localhost:3000** to view the rendered component:



Student Details:

Name: Steeve

School: DAV Public School

Total: 284 Marks

Score: 94.67%

Screenshot: Final output rendered in the browser at localhost:3000

End of Lab